11. A linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ first reflects points through the $x_1$-axis and then reflects points through the $x_2$-axis. Show that T can also be described as a linear transformation that rotates points about the origin. What is the angle of that rotation?

$\boxed{\text{Solution:}}$ $\boxed{\text{Transformation 1:}}$ $T(e_1)=e_1$; $T(e_2)=-e_2$

$\boxed{\text{Transformation 2:}}$ $T(e_1)=-e_1$, $T(-e_2)=-e_2$.

By Theorem 10, the standard matrix is $\begin{bmatrix} T(T(e_1)) & T(T(e_2)) \end{bmatrix}$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} \cos\pi & -\sin\pi \\ \sin\pi & \cos\pi \end{bmatrix}$$

$\therefore$ T is a transformation that rotates points $\pi$ radians anti-clockwise. $\square$

12. Show that the transformation in Exercise 8 is merely a rotation about the origin. What is the angle of the rotation?

$\boxed{\text{Solution:}}$ $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \cos\frac{3\pi}{2} & \sin\frac{3\pi}{2} \\ -\sin\frac{3\pi}{2} & \cos\frac{3\pi}{2} \end{bmatrix}$

$\therefore$ In Exercise 8, T was a transformation that rotates points $\frac{3\pi}{2}$ radians clockwise.

15. $\begin{bmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3x_1-2x_3 \\ 4x_1 \\ x_1-x_2+x_3 \end{bmatrix}$

Fill in the missing entries of the matrix, assuming that the equation holds for all values of the variables.

$\boxed{\text{Solution:}}$ $\begin{bmatrix} 3x_1-2x_3 \\ 4x_1 \\ x_1-x_2+x_3 \end{bmatrix} = \begin{bmatrix} 3x_1 \\ 4x_1 \\ x_1 \end{bmatrix} + \begin{bmatrix} 0x_2 \\ 0x_2 \\ -x_2 \end{bmatrix} + \begin{bmatrix} -2x_3 \\ 0x_3 \\ x_3 \end{bmatrix}$ (vector addition)

$= x_1\begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} + x_2\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} + x_3\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$ (Scalar multiplication)

$= \begin{bmatrix} 3 & 0 & -2 \\ 4 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ (By defn. of matrix equation $Ax=b$)